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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

AD2202-1- BUSINESS INTELLIGENCE
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is semi-structured data?**
A) Data organized in a database
B) Data that has no identifiable structure
C) Data that has some organizational properties but is not fully structured
D) Data with clear relations and table format
2. **OLTP systems are primarily used for which of the following?**
A) Reporting and analytics
B) Transaction processing
C) Data mining
D) Historical data analysis
3. **What does the term "data mart" refer to in BI?**
A) A small data warehouse tailored for specific needs
B) A store that sells data
C) A place where raw data is stored
D) A large collection of unstructured data
4. **What is the primary function of the ETL process?**
A) Analyze historical data
B) Extract, clean, and integrate data from multiple sources into a data warehouse
C) Manage transactional data in real time
D) Backup data from the database
5. **What is the purpose of a data warehouse in BI?**
A) To store real-time transactional data
B) To store large amounts of structured and historical data for analysis
C) To delete unnecessary data
D) To handle real-time analytics
6. **Who is BI typically for?**
A) Data entry clerks
B) Consumers
C) Only IT specialists
D) Decision-makers and business analysts
7. **Which of the following is NOT a characteristic of BI?**
A) Helps in decision-making
B) Provides historical data for analysis
C) Provides real-time transactional data
D) Extracts meaningful insights from raw data
8. **Which of the following is a common approach for building a data warehouse?**
A) ETL (Extract, Transform, Load)
B) Data cleansing
C) Data migration
D) Database normalization
9. **What does data quality primarily refer to?**
A) Data storage methods
B) Accuracy, completeness, and reliability of data
C) Data size and format
D) Data transmission speed



10. Which of the following is NOT a key component of data quality?
A) Accuracy
B) Completeness
C) Reliability
D) Transmission speed
11. What is a multidimensional data model?
A) A model that captures data in a flat structure
B) A model that organizes data into multiple dimensions for better analysis
C) A model used for transactional processing
D) A relational database model
12. Which of the following is an example of a fact in data modelling?
A) A product name
B) A customer's address
C) A region
D) A transaction amount
13. Which of the following is a step in the dimensional model lifecycle?
A) Design of fact tables and dimensions
B) Normalization of relational databases
C) Real-time transactional data analysis
D) Data encryption
14. Which of the following is a key characteristic of an effective dashboard?
A) Shows only historical data
B) Provides real-time, easily interpretable visualizations
C) Contains unstructured data
D) Shows all available data without summarizing
15. What does the SMART test stand for in relation to metrics?
A) Simple, Manageable, Accurate, Reliable, Time-bound
B) Specific, Measurable, Achievable, Relevant, Time-bound
C) Structured, Measurable, Analysable, Relational, Time-sensitive
D) Simple, Multi-dimensional, Accurate, Real-time, Testable
16. Which of the following describes a Key Performance Indicator
A) A qualitative metric used to analyse social media data
B) A quantifiable measure used to evaluate the success of an organization in reaching objectives
C) A type of data warehouse
D) A database normalization technique
17. What is clustering in data mining?
A) Finding distinct groups in the data
B) Classifying data points
C) Finding the relationship between datasets
D) Predicting future trends
18. Which pattern type is focused on identifying relationships between variables?
A) Classification
B) Association
C) Clustering
D) Regression
19. Which of the following is true for the Apriori algorithm?
A) It generates candidate itemsets in a top-down manner
B) It works best for small datasets
C) It ignores support and confidence measures
D) It is primarily used for clustering
20. Which of the following is the correct process for generating association rules?
A) Find all frequent itemsets, then calculate support and confidence for rules
B) Cluster data, then generate rules from clusters
C) Predict future itemsets based on current rules
D) Use decision trees to generate rules

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Apply the concept of structured, semi-structured, and unstructured data to classify data generated by a hospital management system.	8	L*3	1	1
	b) Break down a BI framework into its four major components and discuss how each contributes to the overall decision-making process in an enterprise.	8	L2	2	1
2.	a) Given a retail company's sales data, demonstrate how OLTP and OLAP systems are used for data handling.	8	L3	1	1
	b) Apply the two data integration techniques to combine data from two different hospital departments into a unified system.	8	L3	2	1
3.	a) Compare two different approaches for building a data warehouse by applying them to a university's student information system.	8	L3	2	1
	b) What is BI? Describe who benefits from using business intelligence applications within and outside of an organization.	8	L2	1	1
Unit – II					
4.	a) Apply the concept of data profiling to identify data quality issues in a company's customer contact database.	8	L3	4	1
	b) Explain the concept of star schema and snowflake schema with a suitable example.	8	L2	3	1
5.	a) Given a sales dataset, demonstrate how you would design a multidimensional data model to analyze sales across region, product, and time.	8	L3	3	1
	b) Explain the concepts of dashboard and balanced score cards.	8	L2	4	1
6.	a) Using the concept of dimensions and facts, illustrate how you would model data for a hospital management system.	8	L3	3	1
	b) What is a metric. Explain the SMART test to evaluate a metric.	8	L2	4	1
Unit – III					
7.	a) Consider a frequent itemsets from a market basket dataset, apply the method of generating association rules to determine which rules meet a minimum confidence threshold of 70%.	8	L3	5	1
	b) Explain the terms "support" and "confidence" in association rule mining.	4	L2	5	1
	c) Explain the steps involved in data mining.	4	L2	5	1
8.	a) Given a retail transaction dataset, apply data mining functionalities to demonstrate which functionality (classification, clustering, prediction, association, or outlier analysis) would be most suitable to analyze customer buying behavior.	8	L3	5	1
	b) Describe the steps involved in the Apriori algorithm.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
